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Development of Room Temperature Curable Natural Polyphenols-Based Hybrid Epoxy Polymers

Shinji Kanehashi^{*1,#}, Hiroki Oyagi^{*2}, Kenji Ogino^{*3}, and Tetsuo Miyakoshi^{*2,#}

^{*1}Graduate School of Engineering, Tokyo University of Agriculture and Technology,
2-24-16 Nakacho, Koganei-shi, Tokyo 184-8588, Japan

^{*2}Department of Applied Chemistry, Meiji University, 1-1-1 Higashi-mita, Tama-ku, Kawasaki 214-8571, Japan

^{*3}Graduate School of Bio-Applications Systems Engineering, Tokyo University of Agriculture
and Technology, 2-24-16 Nakacho, Koganei-shi, Tokyo 184-8588, Japan

Abstract: Novel solvent-free bio-based hybrid coating and resin were prepared at room temperature from natural polyphenol compounds, natural lacquer (*i.e.*, urushiol), and cashew nut shell liquid (CNSL). The bio-based hybrids were fabricated by sol-gel and crosslink reactions among organic silane, urushiol, and CNSL-based epoxy compounds. The drying, hardness, and physical properties of the bio-based hybrids were investigated to determine the effective preparation condition. Prepolymerization of epoxy cardanol and epoxy CNSL improved the molecular weight, drying, and hardness properties of the hybrids. The drying time until harden dry (HD) of the hybrids took about 1.3-6.0 hours at room temperature, which is faster than that of the hybrids prepared with commercial bisphenol A-type epoxy (BPAE). The hardness of the hybrids was greater than that of the hybrids prepared with BPAE. Therefore, the epoxy CNSL prepolymers can be effective to prepare the bio-based coating and resin materials compared with the BPAE-based hybrids.

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1. Introduction

Polyphenols are widely distributed in natural resources such as fruits, leaves, and wood [1]. These natural polyphenols are used as antioxidants and radical-scavengers because these natural polyphenols capture and quench free radicals [2-7]. Among polyphenols, natural lacquers (*e.g.*, urushiol) and cashew nut shell liquid (CNSL) are non-food natural polyphenol compounds. Interestingly, the chemical structure of CNSL has a long unsaturated alkyl chain which is similar structure of urushiol [8-11].

The natural lacquer is one of the traditional natural crosslinkable polymers in Japan and can cure naturally based on enzymatic polymerization and autoxidation under the particular temperature and humidity. Once the coating dries completely, it is not soluble in any solvent because of the complicated crosslink structure. Traditionally, the natural lacquer has been used as a coating material because the

lacquer confers specific features such as beautiful gloss, long-term stability, and chemical resistance. Functional properties and thermal treatment of urushiol have been reported [12, 13]. Recently, this urushiol has been investigated as application to functional materials such as nano-coatings [14-16]. Urushiol is now important materials for not only traditional coating but also functional materials. However, the curing of natural lacquer generally takes longer than that of conventional organic coatings because these reactions progress slowly. Furthermore, it is difficult to fabricate thick resin products from natural lacquer alone. Therefore, fabrication of thick resin material using natural lacquer is important to expand the further applications of natural lacquer for effective utilization of renewable resources. We previously reported urushiol-based hybrids that can cure at room temperature [17]. The urushiol-based resins are fabricated by sol-gel and crosslink reactions between

correspondence to Shinji Kanehashi and Tetsuo Miyakoshi

Tel: +81-42-388-7212, Fax: +81-42-388-7404, E-mail: kanehasi@cc.tuat.ac.jp, miya@meiji.ac.jp

organic silane and natural lacquer. The resins are good self-standing with a certain thickness (e.g., 1 cm³ cubic). Using these materials, we have also developed composite materials containing various magnetic metals and carbons for application of microwave absorption [18]. These composites can maintain significantly high loading of fillers due to the binding effect of natural lacquer property. However, these composites can be prepared by oil-based conventional epoxy compound, bisphenol A epoxy (BPAE). Therefore, one of the motivation of this work is that we attempt to prepare the novel bio-based hybrids without use of BPAE.

CNSL is also important raw material for industrial applications such as coating, resin, adhesive, and automobile parts [19, 20]. However, these products have been fabricated with formaldehyde, strong acids (e.g., sulfuric acid), and heavy metals. From an environmental concern, simple fabrication process would be required and important without such toxic chemicals. Recently, we have been focusing on CNSL-based epoxy polymers for resins and coatings [21, 22]. These CNSL-based epoxy polymers were obtained from epoxy CNSL and amine compounds. Therefore, we would expect that this epoxy CNSL can function as an alternative compound of oil-based epoxy compounds.

Here, we investigated the simple preparation of novel bio-based hybrids that can cure at room temperature without any solvent, metal catalyst, or thermal treatment, from natural polyphenols as an alternative compound of oil-based bisphenol compound for the point of green chemistry. The molecular weight, drying, hardness, gel content, and thermal properties of the bio-based hybrids were characterized.

2. Experimental

2.1 Chemicals

The chemical structures of key compounds used in this work are presented in Figure 1. As renewable resources, natural lacquer was purchased from Doityu Shoten, Co. Ltd., Osaka, Japan, and CNSL was kindly provided by Cashew Co., Ltd., Saitama, Japan. We obtained urushiol from this natural lacquer according to previous work [10]. CNSL is a mixture of phenolic compounds with cardanol as the major component and with cardol and 2-methylcardol as minor components. The composition of the CNSL

used in this work was reported elsewhere [22]. Epoxy cardanol (EC) and epoxy CNSL (ECN) were obtained from epoxidation of cardanol and CNSL [23, 24]. The prepolymerization of EC and ECN was based on autoxidative polymerization by thermal treatment [21, 22]. In this study, we used the prepolymers prepared at 140 °C for 24 hours. An organic silane compound, *N*-(2-aminomethyl)-3-aminopropyl trimethoxysilane (AATMS), was purchased from Tokyo Chemical Industry Co., Ltd., Tokyo, Japan. These chemicals were used without further purification.

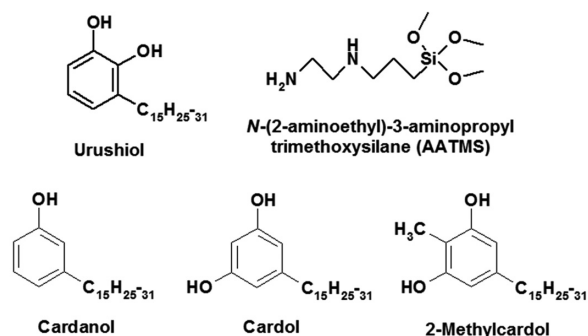


Fig. 1 Chemical structure of natural polyphenols and organosilicon compounds.

2.2 Preparation of Natural Polyphenol-based Hybrids

Figure 2 shows the preparation protocol for natural polyphenol-based hybrids. 10 grams of urushiol and given ratios of additives such as EC, ECN, or their prepolymers (i.e., ECP and ECNP) and organosilicon compounds were mixed for 5 min at room temperature. For Entry 3 listed in Table 1, 10 g of urushiol, 0.5 g of AATMS as organosilicon compound, and 0.5 g of epoxy compound were mixed. After mixing these compounds well, the mixture was degassed and uniformly coated at 76 μm thick on a square glass plate (70 mm×70 mm×1.3 mm) at 23 °C±1 °C using an applicator (Yoshimitsu Seiki, Tokyo, Japan) which has ±10 μm deviation. This lacquer was stored in the dark at 25 °C and 50% relative humidity to evaluate the drying and hardness properties. For preparation of the thick resin, the mixture of natural lacquer, ECP or ECNP, and organosilicon compounds was poured into a fluororesin tube (outside diameter: 16 mm, inside: 14 mm) at 23 °C±1 °C. After 3 days, the thick resin that had formed was taken out of the tube.

2.3 Structure Analysis

All characterization data were obtained using a minimum of three samples to confirm the reproducibility of the experimental results. ¹H NMR spectroscopy was conducted on a JNM-ECA 500 spectrometer (JEOL Ltd., Tokyo, Japan). Samples

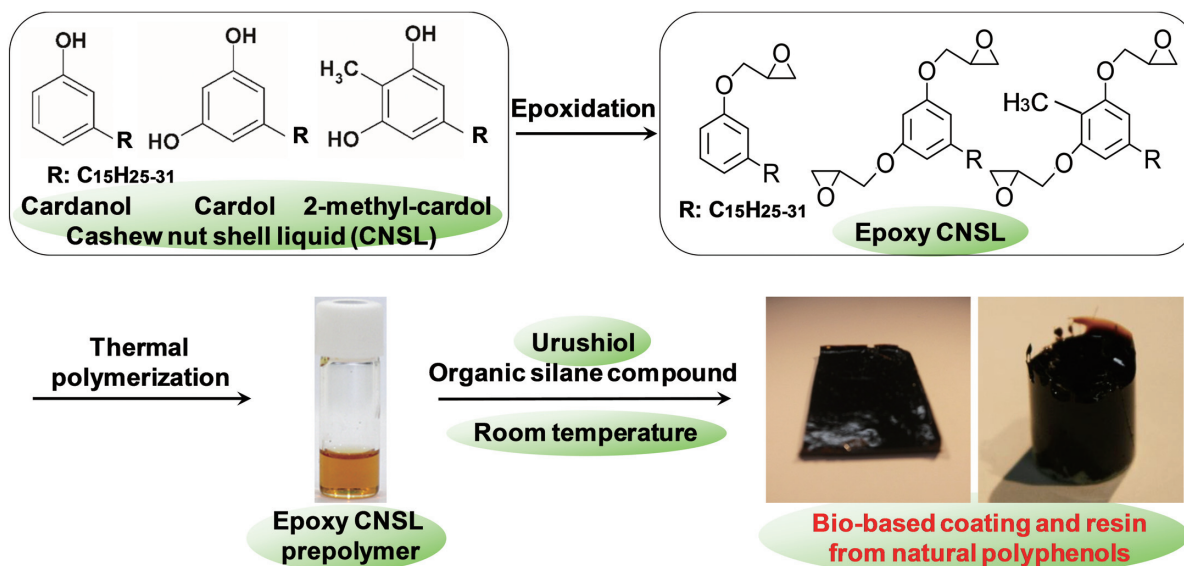


Fig. 2 Preparation of natural polyphenol-based hybrids.

were dissolved in deuterated dimethyl sulfoxide (DMSO) with chemical shifts referenced from tetramethylsilane (TMS). The molecular weight of ECP was determined at 40 °C by gel permeation chromatography (GPC; TSK-gel column α -3000, α -4000 and α -M, ϕ 7.8 mm $\times$ 300 mm $\times$ 3, Tosoh Co. Ltd., Tokyo, Japan) using dimethylformamide (DMF) as an eluent with 0.01 mol LiBr on a high-performance liquid chromatography system with a refractive-index detector RI-8012 with polystyrene standards. The elution rate was 0.8 mL/min. In the present work, oligomer expresses higher molecular weight of monomer and less than 10,000 g/mol (*i.e.*, molecular weight of polymer).

2.4 Characterization

We focused on the properties of coating in terms of drying, hardness, thermal, and mechanical properties. These properties are also fundamental and important properties as coating and resins materials. The drying process of the bio-based coatings at 23 °C can be divided into three stages: dust-free dry (DF), touch-free dry (TF), and hardened dry (HD). DF is that the coating has hardened sufficiently to prevent dust from becoming embedded in the coating. TF is that the coating can be touched by finger, while HD is that the coating is hard enough. Each stage was measured using an automatic drying time recorder (RC auto-recorder of paint drying time, TaiYu Co. Ltd., Osaka, Japan) at 23 °C and 60% relative humidity. The hardness of pencil test is determined using designations consisting of letters and numbers according to the current national standard of GB/T 6739-1996. The meanings of H and B designate how

hard or soft the tested coatings are, respectively. A higher number with H or B expresses the hardness or softness of the tested coatings, respectively. F and HB indicate medium hardness. F is a slightly harder coating than HB. In the present study, pencil lead hardness was determined using a C-221 (Yoshimitsu Seiki, Tokyo, Japan) at 23 °C. The gel content of the bio-based coating was determined. Coatings were immersed in acetone at 23 °C for 24 h, and the non-soluble parts were filtered and dried in a Taiyo muffle furnace (Isuzu, Tokyo, Japan) overnight at 50 °C to remove residual solvent, cooled, and subsequently weighed at room temperature. The gel content was calculated based on the weight of the insoluble fraction and the original weight of the completely dried hybrids. The coating chromaticity was determined with a Spectro-Guide 45 / 0 spectrophotometric colorimeter (BYK Gardner GmbH, Germany) with a white background based on ISO 11664. Thermogravimetric analysis (TGA) was performed with a DTG-60 (Shimadzu Corporation, Japan) from 50 °C to 600 °C in a platinum pan at a heating rate of 10 °C/min and a flow rate of 60 mL/min under a nitrogen atmosphere. A rigid-body pendulum physical property-testing (RPT), RPT-3000 W (A&D Ltd., Tokyo, Japan) was used to determine the viscoelasticity which is related with crosslink density of hybrids. The glass transition temperature, T_g , of the hybrid was also analyzed by RPT. The heat of the RPT oven was programmed to rise at a rate of 4.3 °C/min from -35 °C to 200 °C.

3. Results and discussion

3.1 Characterization

Table 1 summarizes the molecular weight distribution of natural polyphenol-based hybrids prepared with 5 wt% AATMS. As the content of EC, ECN, and their prepolymers increased, the polymer content, molecular weight (M_w), and polydispersity index, PDI (M_w/M_n) also increased. It is expected that the molecular weights of natural polyphenol-based hybrids prepared with the prepolymers were higher than those prepared with EC or ECN monomer. On the other hand, molecular weights of hybrids prepared with ECN were higher than those prepared with EC. This is because ECN is a mixture of epoxy cardanol, epoxy cardol, and epoxy 2-methylcardol, indicating that ECN has more epoxy groups, that is, crosslinking points in the structure. These trends were well consistent with the diamine-ECNP reaction reported previously [22]. As compared with previous hybrid prepared with 5 wt% BPAE and 5 wt% AATMS, the polymer content of these hybrids in this work was lower than that in the previous hybrid, while the monomer content was higher than that in the previous hybrid [17]. Highly monomer content means that there are some unreacted monomers in

the hybrids, indicating that lower polymer content is expected. Therefore, the reactivity among these epoxy, amino silane, and urushiol compounds was lower than those of previous hybrid systems. In contrast, the molecular weights of hybrids prepared with prepolymers were greater than that of previous BPAE. This means that prepolymerization can provide the higher molecular weight of hybrids prepared from EC or ECN prepolymers, regardless of the content of monomers, oligomers, and polymers as shown in Table 1.

Table 2 summarizes the drying and hardness properties of the natural polyphenol-based hybrids. As compared with natural lacquer (Entry 1) and previous hybrid (Entry 18), the natural polyphenol-based hybrids significantly improved the drying property at room temperature. In addition, the times until HD was reached for these hybrids were significantly shorter than the previous hybrid prepared with 5 wt% BPAE and 5 wt% AATMS. Reactivity among epoxy, amine, and urushiol compounds affect the drying property. Once molecular weight of hybrids become higher, the hybrids can cure. Therefore, this rapid drying property of these hybrids suggests that initial reaction between epoxy, amine, and urushiol

Table 1 Characteristics of natural polyphenol-based hybrids (AATMS : 5 wt%)

Entry	Epoxy	Epoxy content (wt%)	Content ratio (%) ^(a)			Molecular weight		
			Monomer	Oligomer	Polymer	M_n	M_w	M_w/M_n
1 ^(b)	None	0	49.1	45.7	5.2	510	2140	4.2
2	EC	5	44.2	43.0	12.8	470	4040	8.6
3	EC	10	47.2	41.7	11.2	530	3950	7.5
4	EC	15	47.0	42.1	10.9	540	6100	11.3
5	EC	20	46.9	41.3	11.0	400	5460	13.7
6	ECP	5	48.2	40.4	11.4	520	9800	18.8
7	ECP	10	41.9	44.7	13.4	680	4140	6.1
8	ECP	15	42.8	43.8	13.4	480	4550	9.5
9	ECP	20	40.9	46.3	12.9	360	8550	23.8
10	ECN	5	44.2	43.0	12.8	470	4040	8.6
11	ECN	10	43.5	44.0	12.5	470	4960	10.6
12	ECN	15	45.8	42.9	11.3	490	6250	12.8
13	ECN	20	44.2	41.8	14.0	360	4530	12.6
14	ECNP	5	42.7	43.5	13.8	490	6100	12.4
15	ECNP	10	42.6	43.6	13.8	570	9120	16.0
16	ECNP	15	42.4	43.8	13.8	480	10600	22.1
17	ECNP	20	43.8	41.7	14.5	480	12400	25.8
18 ^(b)	BPAE	20	24.4	48.4	27.2	1070	7560	7.1

(a) Oligomer: monomer $\leq$ molecular weight < 10,000 g/mol, polymer: molecular weight $\geq$ 10,000 g/mol

(b) Cited from [17]

Table 2 Drying property and hardness of natural polyphenol-based hybrid coatings (AATMS: 5 wt%)

Entry	Epoxy	Epoxy content (wt%)	Drying property (h) ^(a)			Hardness ^(b)					
			DF	TF	HD	1 day	3 days	5 days	7 days	10 days	15 days
1 ^(c)	None	0	18.5	23.1	24 <	TF	HD	HD	HD	–	6B
2	EC	5	0.4	1.0	2.8	6B	3 B	H	H	2H	4H
3	EC	10	0.5	1.2	4.1	5B	B	2H	3H	3H	4H
4	EC	15	0.5	1.2	4.0	6B	4 B	HB	F	H	2H
5	EC	20	0.6	1.4	4.3	6B>	6B>	6B>	6B>	6B>	6B
6	ECP	5	0.4	1.2	1.3	4B	HB	2H	2H	3H	6H
7	ECP	10	0.4	1.3	1.6	5B	HB	2H	3H	4H	5H
8	ECP	15	0.5	1.2	2.2	5B	2B	HB	H	2H	3H
9	ECP	20	0.5	1.2	2.4	6B	3B	F	F	H	H
10	ECN	5	0.5	1.0	2.6	5B	B	2H	2H	3H	6H
11	ECN	10	0.5	1.1	2.6	5B	2B	H	2H	3H	6H
12	ECN	15	0.5	1.1	4.1	5B	3B	HB	F	H	3H
13	ECN	20	0.6	1.2	6.0	6B	5B	4B	B	HB	F
14	ECNP	5	0.5	1.1	2.1	4B	F	3H	4H	4H	6H
15	ECNP	10	0.5	1.1	2.4	5B	B	2H	3H	3H	6H
16	ECNP	15	0.5	1.2	4.0	6B	3B	F	H	2H	5H
17	ECNP	20	0.6	1.3	4.3	6B	3B	F	H	H	3H
18 ^(b)	BPAE	20	0.9	4.6	13.1	HD	6B	–	6B	–	4B

(a) DF: dust free dry, TF: touch free dry, and HD: harden dry (based on JIS-K-5400)

(b) Pencil hardness: HD << 6B < B < HB < F < H << 9H

(c) Cited from [17]

proceeded effectively. On the other hand, the natural polyphenol-based hybrids were 6 H at the maximum which was harder than the previous hybrid (Entry 18) as presented in Table 2. Based on these results, EC, ECN, and their prepolymers improved the drying and hardness properties. As the epoxy content increased, the hybrids dried more slowly and became more flexible, suggesting that the crosslink structure became more complicated. The previous CNSL-based epoxy polymers, the hardness was 3 B after 2 weeks [22]. Prepolymerization of EC and ECP provided more rapid drying and higher hardness. As shown in Table 1, the drying and hardness of natural polyphenol-based hybrids were also improved by the high molecular weight of EC and ECN prepolymers. Therefore, we evaluated the molding property of hybrids prepared with ECNP.

Figure 3 shows the gel content of the natural polyphenol-based hybrids prepared with ECN or ECNP and 5 wt% AATMS. The gel content of natural lacquer increased with time, indicating that typical autoxidation still proceeded. The gel content was increasing even after three weeks. By contrast, the gel content of natural polyphenol-based hybrids prepared with 5 or 10 wt% ECN or ECNP ranged from 80 to 87 wt%. The gel content of all hybrids

initially increased based on sol-gel reaction until 1 day, then slightly increased based on autoxidative after 1 day. Thus, the hybrids cure rapidly compared with natural lacquer. These results also suggest that the crosslink densities of these hybrids were higher than that of natural lacquer. However, the gel content of these hybrids were lower than those of previous hybrids prepared with 5 wt% of BPAE and 5 wt% of AATMS, suggesting that the reactivity of ECN and ECNP was less than that of BPAE, as we can expect

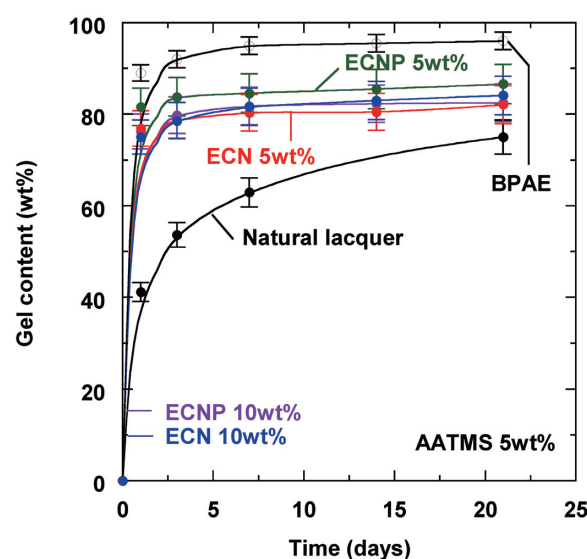


Fig. 3 Gel content of natural polyphenol-based hybrids.

Table 3 Color property of natural polyphenol-based hybrid coatings (AATMS content: 5 wt%)

Entry	Epoxy	Epoxy content (wt%)	Color indices		
			L^*	a^*	b^*
1	None	0	11.8	29.5	17.0
2	ECN	5	1.7	0.9	0.5
3	ECN	10	2.4	1.5	0.5
4	ECN	15	3.7	4.7	1.3
5	ECN	20	2.8	2.7	1.0
6	ECNP	5	2.0	1.6	0.7
7	ECNP	10	2.1	2.9	1.0
8	ECNP	15	2.9	2.9	0.9
9	ECNP	20	2.8	3.1	0.9
10	BPAE	20	2.0	7.6	2.5

from Table 1.

Table 3 summarizes the chromaticity of natural polyphenol-based hybrids prepared with 5 wt% AATMS. The coating was analyzed after 7 days. CIE measurement of UV-cured films was performed to characterize changes in film color at different times after UV irradiation. The film colors were characterized by the CIE Lab system using a colorimeter [25]. Coating brightness index, L^* , varies from 0 to 100, meaning that the maximum, 100, corresponds to perfect brightness, while the minimum, 0, corresponds to perfect blackness. The redness ($a^* > 0$)/greenness ($a^* < 0$), a^* and yellowness ($b^* > 0$)/blueness ($b^* < 0$), b^* indices have no specific numerical limits. The experimental deviation was less 5% considering the sample and measurement errors. At first, the natural lacquer showed a dark color based on the color indices. On the other hand, the brightness, redness, and yellowness of natural polyphenol-based hybrids gradually increased with epoxy content, suggesting that the film colors became brighter and paler. These differences in the color may imply the different crosslink structures.

Figure 4 presents the TGA curves of natural polyphenol-based hybrids. The residual weight of bio-based polymers began to decrease gradually from 100-150 °C owing to the presence of unreacted monomers, moisture, and byproducts (e.g., methanol) after sol-gel reaction. As compared with natural lacquer, better thermal resistance was observed in natural polyphenol-based hybrids prepared with 5 wt% ECN and 5 wt% ECNP as well as the hybrid prepared with 5 wt% BPAE. The hybrids prepared with 10 wt% ECN and 10 wt% ECNP were very similar in thermal decomposition behavior to the

natural lacquer. Based on these results, the relatively high contents of ECN and ECNP reduced the thermal stability of the hybrids. Therefore, the thermal stability of these hybrids prepared with lower content of ECN and ECNP was improved compared with natural lacquer.

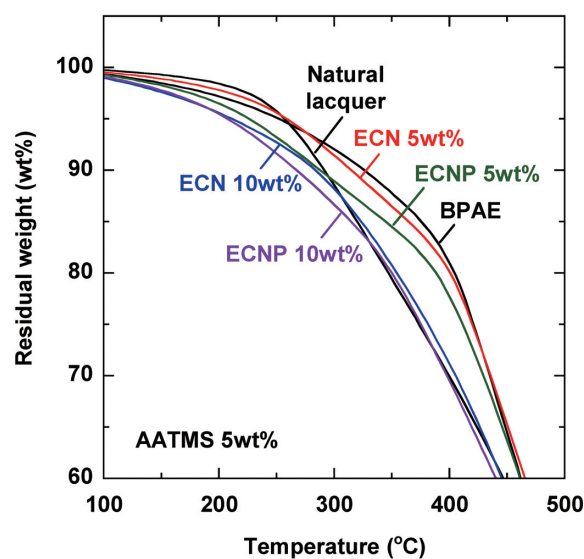


Fig. 4 TGA curves of natural polyphenol-based hybrids.

Figure 5 shows the glass transition temperature of natural polyphenol-based hybrids. T_g was determined by RPT measurement. As we expected from the gel content, the T_g of natural lacquer increased gradually from 69 °C to 124 °C over three weeks due to the slow autoxidation. On the other hand, the maximum T_g value, 194 °C, after three weeks was observed in the hybrid prepared with 5 wt% ECNP. The order of T_g was: 5 wt% ECNP (194 °C) > 5 wt% ECN (184 °C) > 10 wt% ECNP (169 °C) or 10 wt% ECN (172 °C). This result is very consistent with the thermal stability presented in Figure 4. All T_g

were 36% to 57% greater than those of natural lacquer. This is because that the crosslink density of natural polyphenol-based hybrids is higher than that of natural lacquer due to the more complicated rigid crosslink structure as presented in Figure 5.

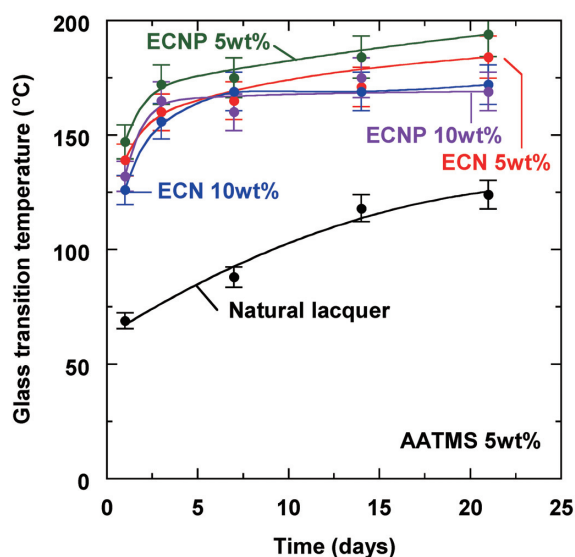


Fig. 5 Glass transition temperature of natural polyphenol-based hybrids.

3.2 Structure Analysis

To analyze the polymerization mechanism and investigate the reactivity of each compound of the natural polyphenol-based hybrids, we measured ^1H NMR of soluble parts of the hybrids. Figure 6 shows ^1H NMR spectra of ECNP and soluble parts of natural

polyphenol-based hybrids (5 wt% ECNP and 5 wt% AATMS) after 1 day and 3 days. Decreases in the peak at 4.25, 3.78, and 2.81–2.69 ppm after crosslink reaction, which corresponds to ECNP, were observed, whereas the peak at 6.7–6.5 ppm, which corresponds to the aromatic proton of urushiol, decreased as compared with raw urushiol, although the degree of decrease was smaller than the previous hybrid prepared with BPAE. A peak at 3.2 ppm, which corresponds to the methanol produced by alcoholysis and the sol-gel reaction between urushiol and AATMS, appeared in the hybrids. As we reported the possible structure of the previous hybrid prepared with BPAE based on GPC and NMR analyses, the sol-gel reaction between urushiol and the organic silane compound proceeded immediately, then the reaction between BPAE and the organic silane progressed slowly after the epoxy ring opening [17]. In the present case in the bio-based epoxy compounds, the results of GPC and NMR analyses suggest that the sol-gel reaction between urushiol and AATMS, an organic silane compound, proceeded; however, this reactivity could be lower than the previous hybrid prepared with BPAE, indicating that the ECNP could inhibit the sol-gel reaction between urushiol and AATMS due to the steric hindrance by ECNP. After that, the epoxy crosslink reaction with ECNP occurred in the hybrids as a similar reaction in the

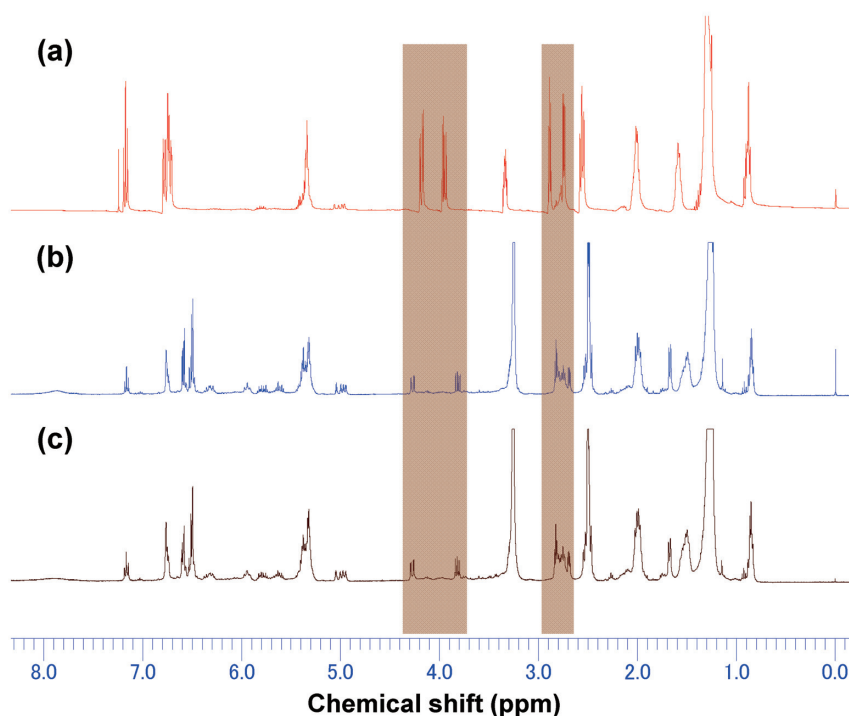


Fig. 6 ^1H NMR spectra of ECNP and soluble parts of natural polyphenol-based hybrids (5 wt% ECNP and AATMS 5 wt%). (a) ECNP, soluble parts of natural polyphenol-based hybrids after 1 day (b) and 3 days (c), respectively.

previous hybrids.

3.3 Moldability

The molding property of the hybrids was evaluated by the appearance of resins, as presented in Figure 7. Table 4 also summarizes the molding property of natural polyphenol-based hybrid resins prepared at room temperature. If a resin can be removed from the tube and keep its shape, it has a good molding property, while a resin that doesn't cure has a poor molding property. First, the natural lacquer didn't cure thick resin. It is well known that the autoxidation of natural lacquer progresses quite slowly under specific conditions (*e.g.*, humidity and temperature), the curing of natural lacquer generally takes longer than other conventional coatings, and such a thick resin cannot cure completely inside [17]. On the other hand, the hybrids prepared with ECN also did not cure completely at room temperature, regardless of the preparation composition. In contrast, we successfully obtained thick hybrid resins prepared with ECNP. The higher molecular weight and better drying and hardness properties significantly improved their molding property. Therefore, we found that ECNP can work as an alternative oil-based BPAE and improved the biomass content of this hybrid.

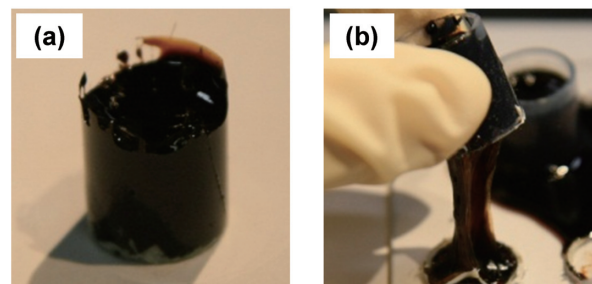


Fig. 7 Appearance of natural polyphenol-based hybrids. (a) Cured and (b) uncured.

4. Conclusions

Novel natural polyphenol-based hybrids were prepared at room temperature from urushiol, CNSL as a renewable material, and organic silane compounds. The drying, hardness, and thermal properties of the hybrids were investigated to optimize the preparation condition (*e.g.*, composition). Prepolymerization of EC and ECN was effective to improve the hybrid properties such as molecular weight, drying, and hardness properties. NMR analysis showed that the sol-gel reaction occurred between natural lacquer and organic silane, and a crosslink reaction between organic silane and epoxy groups proceeded. However, the reactivity of these hybrids was lower than that of previous BPAE, suggesting that ECNP inhibits the sol-gel reaction

Table 4 Moldability of natural polyphenol-based hybrid resins

Entry	Epoxy compound	Additive content (wt%)		Moldability ^(a)
		AATMS	Epoxy	
1 ^(b)	None	0	0	×
2	ECN	20	20	×
3	ECN	20	30	×
4	ECN	20	40	×
5	ECN	30	30	×
6	ECN	30	40	×
7	ECN	40	40	×
8	ECNP	20	20	△
9	ECNP	20	30	○
10	ECNP	20	40	○
11	ECNP	30	30	○
12	ECNP	30	40	○
13	ECNP	40	40	○
14 ^(b)	BPAE	10	30	△
15 ^(b)	BPAE	20	20	○
16 ^(b)	BPAE	30	10	△

(a) ○: cured, △: partially cured but brittle, and ×: uncured

(b) Cited from [17]

between urushiol and AATMS due to the lower reaction at the OH groups of urushiol. Therefore, the epoxy CNSL prepolymers can be used as an alternative compound of oil-based BPAE and effective to prepare the bio-based coating and resin materials.

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